

AI Price Predictor

Predictive Modeling & Agentic Workflow Project for Cleaning Services

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APPLICATION & DATASET SCOPE

Application Description

The Smart Estimator is an intelligent system designed to predict professional cleaning costs in EUR. It leverages a sophisticated agentic workflow and predictive modeling to provide accurate pricing estimates based on natural language or structured inputs.

- Predicts cleaning costs for properties 30-200 m²
- Factors in rooms, bathrooms, pets, and gardens
- Supports weekly, biweekly, and monthly schedules



Dataset Overview

Comprehensive regression dataset containing 1,000 rows and 7 columns with no missing values, used for high-fidelity training.

- **Features:** Area (avg ~115 m²), Rooms (1-8), Bathrooms (1-3)
- **Indicators:** Pets (40.6%), Garden (32.9%)
- **Target:** Price range €18.37 - €112.04 (avg €58.78)



PROJECT ARCHITECTURE: 4 TOOLS + AGENT



1. Prediction Engine

Supervised learning regression model using Model Reader & Predictor nodes to estimate price.



2. Confidence Scorer

Returns a $\pm 10\%$ price range around the predicted value.



3. LLM Parser

Converts natural language input into structured tables using LLM Prompter for seamless UX.



4. Feature Analytics

Calculates feature importance metrics to explain which house attributes impact price most.



The AI Orchestrator (Agent)

A sophisticated loop utilizing "Workflow to Tool" & "Agent Prompter" to route user queries, invoke the correct tools, and return readable conversations.



MODEL RESULTS & DEMO INSIGHT

0.913

Model R² Score



Hello, could you calculate the cleaning cost for a large house? It's 200 square meters, featuring 5 rooms and 3 bathrooms. We also have two dogs and a garden that needs attention. We would like the cleaning to be done weekly.



Thank you for sharing all those details. I've taken note that your home is 200 square meters, has 5 rooms and 3 bathrooms, you have pets (two dogs), a garden, and you'd prefer a weekly cleaning frequency.

The estimated cost for your cleaning service is €150. Since every home is unique, the actual price usually falls somewhere between €130 and €170. Just so you know, the main factors shaping your specific quote are the size of your home, the number of bathrooms, and the fact that you have pets and a garden, which require extra attention, as well as the weekly cleaning frequency.

Does this estimate sound good to you, or do you have any questions?



Hi! I need a price estimate for cleaning my small apartment. It is 45 square meters with just 1 room and 1 bathroom. I don't have any pets and there is no garden. I am looking for a monthly cleaning service.



Thank you for sharing the details about your apartment. I've taken note that it's 45 square meters, has 1 room and 1 bathroom, you don't have pets, no garden, and you're looking for a monthly cleaning service.

The estimated cost for your cleaning service is €40. Since every home is unique, the actual price usually falls somewhere between €35 and €45. Just so you know, the main factors shaping your specific quote are the small size of your apartment and the monthly cleaning frequency.

Does this estimate sound good to you, or do you have any questions?



Hello, could you calculate the cleaning cost for a large house? It's 200 square meters, featuring 5 rooms . We also have two dogs and a garden that needs attention. We would like the cleaning to be done weekly.



You've shared some details about your home, but I just need to know one more thing to give you an accurate estimate. How many bathrooms do you have in your 200 square meter house?